7. Consider the system S characterized by the differential equation

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t)$$

(a) Determine the zero state response for input $x(t) = e^{-4t}u(t)$.

(b) Determine the zero input response of the system for $t > 0^-$, given that

$$y(0^-) = 1, \quad y'(0^-) = -1, \quad y''(0^-) = 1$$

(c) Determine the output of S when the input is $x(t) = e^{-4t}u(t)$ and the initial condition are the same as those specified in part b.

$$H(S) = \frac{1}{(S+1)(S+2)(S+3)}$$

(a) $x(S) = \frac{1}{(S+4)} \Rightarrow Y(S) = \frac{1}{(S+1)(S+2)(S+3)(S+4)}$

$$Y(S) = \frac{1/6}{S+1} + \frac{-1/2}{S+2} + \frac{1/2}{S+3} + \frac{-1/6}{S+4}$$

$$y(t) = \left[1/6\, e^{-t} - 1/2\, e^{-2t} + 1/2\, e^{-3t} - 1/6\, e^{-4t}\right] u(t)$$

(b) $x(t) = 0 \Rightarrow x(S) = 0$

$$\left[S^3Y(S) - S^2\overset{1}{y}(0) - S\overset{-1}{y'}(0) - \overset{1}{y''}(0)\right] + 6\left[S^2Y(S) - S\overset{1}{y}(0) - \overset{-1}{y'}(0)\right]$$

$$+ 11\left[SY(S) - \overset{1}{y}(0)\right] + 6Y(S) = 0$$

$$\Rightarrow Y(S)\left[S^3 + 6S^2 + 11S + 6\right] = S^2 - S + 6S + 1 - 6 + 11 = S^2 + 5S + 6$$

$$\Rightarrow Y(S) = \frac{1}{S+1}$$

$$\Rightarrow y(t) = e^{-t}u(t)$$

(c) Final output = zero state + zero input

$$= \left[7/6\, e^{-t} - 1/2\, e^{-2t} + 1/2\, e^{-3t} - 1/6\, e^{-4t}\right] u(t)$$